



K.R. Mangalam University School of Engineering & Technology

Fundamentals Of Java Programming Lab (ENCA203) Assignment 1 Student Management System

Submitted by:

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Class: BCA (h) (AI & DS)

Submitted to:

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GitHub Repository:

https://github.com/shivamgaur1255-sketch/java_assignments-

Codes:

```
J Calculator.java 1 J StudentRecordSystem.java X StudentApp
J StudentRecordSystem.java > Student > inputDetails(Scanner)
1 import java.util.ArrayList;
2 import java.util.Scanner;
3
4 class Person {
5     String name;
6 }
7
8 class Student extends Person {
9     int rollNo;
10    String course;
11    double marks;
12    char grade;
13
14    Student() {}
15
16    Student(int rollNo, String name, String course, double marks) {
17        this.rollNo = rollNo;
18        this.name = name;
19        this.course = course;
20        this.marks = marks;
21        calculateGrade();
22    }
23
24    void inputDetails(Scanner sc) {
25        System.out.print(s:"Enter Roll No: ");
26        rollNo = sc.nextInt();
27        sc.nextLine();
28
29        System.out.print(s:"Enter Name: ");
30        name = sc.nextLine();
31
32        System.out.print(s:"Enter Course: ");
33        course = sc.nextLine();
34
35        while (true) {
36            System.out.print(s:"Enter Marks: ");
37            marks = sc.nextDouble();
38            if (marks >= 0 && marks <= 100)
39                break;
40            else
41                System.out.println(x:"Invalid marks! Enter between 0-100.");
42        }
43
44        calculateGrade();
45    }
}
```

```
J Calculator.java 1 J StudentRecordSystem.java X StudentApp
J StudentRecordSystem.java > Student > inputDetails(Scanner)
8 class Student extends Person {
24 void inputDetails(Scanner sc) {
45 }
46
47 void calculateGrade() {
48     if (marks >= 90) grade = 'A';
49     else if (marks >= 75) grade = 'B';
50     else if (marks >= 50) grade = 'C';
51     else grade = 'D';
52 }
53
54 void displayDetails() {
55     System.out.println("Roll No: " + rollNo);
56     System.out.println("Name: " + name);
57     System.out.println("Course: " + course);
58     System.out.println("Marks: " + marks);
59     System.out.println("Grade: " + grade);
60     System.out.println(x:"-----");
61 }
62 }
63
64 public class StudentRecordSystem {
65     Run | Debug
66     public static void main(String[] args) {
67         Scanner sc = new Scanner(System.in);
68         ArrayList<Student> students = new ArrayList<>();
69
70         while (true) {
71             System.out.println(x:"\n==== Student Record Menu =====");
72             System.out.println(x:"1. Add Student");
73             System.out.println(x:"2. Display All Students");
74             System.out.println(x:"3. Exit");
75             System.out.print(s:"Enter your choice: ");
76             int choice = sc.nextInt();
77
78             switch (choice) {
79                 case 1:
80                     Student s = new Student();
81                     s.inputDetails(sc);
82                     students.add(s);
83                     break;
84                 case 2:
85                     if (students.isEmpty()) {
86                         System.out.println(x:"No records found!");
87                     }
88             }
89         }
90     }
91 }
}
```

```
J Calculator.java | J StudentRecordSystem.java X | StudentApp
J StudentRecordSystem.java > Student > inputDetails(Scanner)
64 public class StudentRecordSystem {
65     public static void main(String[] args) {
66         while (true) {
67             System.out.println("\n==== Student Record Menu =====");
68             System.out.println("1. Add Student");
69             System.out.println("2. Display All Students");
70             System.out.println("3. Exit");
71             System.out.print("Enter your choice: ");
72             int choice = sc.nextInt();
73
74             switch (choice) {
75                 case 1:
76                     Student s = new Student();
77                     s.inputDetails(sc);
78                     students.add(s);
79                     break;
80                 case 2:
81                     if (students.isEmpty()) {
82                         System.out.println("No records found!");
83                     } else {
84                         for (Student st : students) {
85                             st.displayDetails();
86                         }
87                     }
88                     break;
89                 case 3:
90                     System.out.println("Exiting the application. Goodbye!");
91                     sc.close();
92                     return;
93                 default:
94                     System.out.println("Invalid choice! Try again.");
95             }
96         }
97     }
98 }
99
100
101
102
103
104
105
```

Output:

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

StudentApp

PS C:\Users\Shivam Gaur\OneDrive\ドキュメント> java coding> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Shivam Gaur\AppData\Roaming\Code\User\workspaceStorage\b86f09e45bbf9afd5a31de64f223dfa7\redhat.java\jdt_ws\java_coding_d70f91f4\bin' 'StudentRecordSystem'

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 1
Enter Roll No: 101
Enter Name: Sita
Enter Course: BCA
Enter Marks: 50

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 1
Enter Roll No: 102
Enter Name: Gita
Enter Course: BSC
Enter Marks: 80

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 2
Roll No: 101
Name: Sita
Course: BCA
Marks: 50.0
Grade: C

Roll No: 102
Name: Gita
Course: BSC
Marks: 80.0
Grade: B

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Course: BSC
Marks: 80.0
Grade: B

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Course: BSC
Marks: 80.0
Grade: B

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Course: BSC
Marks: 80.0
Grade: B

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Course: BSC
Marks: 80.0
Grade: B

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Course: BSC
Marks: 80.0
Grade: B

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Course: BSC
Marks: 80.0
Grade: B

===== Student Record Menu =====
1. Add Student
2. Display All Students
3. Exit
Enter your choice: 3
Course: BSC
Marks: 80.0
Grade: B

Exiting the application. Goodbye!
PS C:\Users\Shivam Gaur\OneDrive\ドキュメント> java coding>

Explanation:

There are two classes

- Person → just stores the name
- Student → extends Person and adds roll number, course, marks, and grade

When you add a student

- The program asks you to enter details using the keyboard (Scanner)
- It saves those details inside a Student object
- It checks that marks are between **0–100**
- Based on the marks, it automatically decides the grade (A, B, C, or D)

Where the data goes

- Every student is stored inside an ArrayList
- Think of it like a growing list where each new student is added

Displaying students

- When you choose option **2**
- The program loops through the ArrayList
- It prints each student's details one by one

Menu keeps running

- The program shows the menu again and again inside a loop
- It only stops when you choose **3 (Exit)**

What the program teaches

- Object-oriented concepts (classes, objects, inheritance)
- Using methods to organize tasks
- Taking input and displaying output
- Using conditions and loops
- Storing multiple records using an ArrayList